

Rapidly Build and Deploy a Full-Stack App with Cursor

Advanced AI-assisted SaaS application development

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Cursor 3 & Context Management

AI-First Code Editor for Production Development

Not just autocomplete—full stack development

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Not just autocomplete—full stack development

Composer 2.5

Standard + Fast tiers—an iteration over Composer 2

Agents Window

Parallel agents across local, worktree, cloud & SSH

Bugbot

Automated PR review on Composer 2.5—~90-second reviews

MCP Integration

Connect to GitHub, Figma, databases

[1][Cursor 3 Changelog](#)

Core Features

Chat (Cmd+L)

- Multi-tab conversations
- Context checkpoints
- Quick Q&A

Tab Autocomplete

- Multi-line predictions
- Context-aware
- Pattern learning

Agent (Cmd+I)

- Multi-file editing
- Autonomous execution
- Planning + building

Inline Edit (Cmd+K)

- In-place modifications
- Natural language
- Stay in flow

The Three-Step Framework



1. EXPLORE

Share context
Identify files
Discuss options



2. PLAN

Step-by-step plan
Markdown checklists
Break down tasks



3. BUILD

Execute iteratively
Review each step
Commit regularly

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This framework prevents context overload and ensures quality output

[1] [Cursor for Staff Engineers](#)

Context Management: The Foundation



Key Challenge

Context management is like telling a story—when it gets convoluted, the AI loses track

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Context management is like telling a story—when it gets convoluted, the AI loses track

Why It Matters

- Fixed context windows (128k-1M tokens)
- Comprehension $\neq$ token count
- Poor context = hallucinations

Best Practices

- Use @-mentions for precision
- Start new chats frequently
- Configure `.cursorignore`

The @ Symbol: Your Context Navigator



```
@src/components/Header.tsx
```

Reference specific files



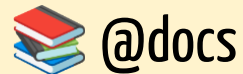
```
@src/utils/
```

Include entire directories



```
@codebase "authentication"
```

Semantic search



```
@docs "React hooks"
```

Query documentation

Effective Prompting



Make this better

- No success criteria
- No direction
- Ambiguous goal



@src/components/Button.tsx Refactor to: 1. Add TypeScript props 2. Include disabled state 3. Add loading spinner 4. Keyboard accessible



Describe the diff you want: current state → desired state

Project Rules: .cursor/rules

Four Application Modes

1. **Always Apply** - Every session automatically
2. **Apply Intelligently** - AI determines relevance (recommended)
3. **Apply to Specific Files** - Pattern matching (*.test.ts)
4. **Apply Manually** - Activated with @-mention

Project Rules: .cursor/rules

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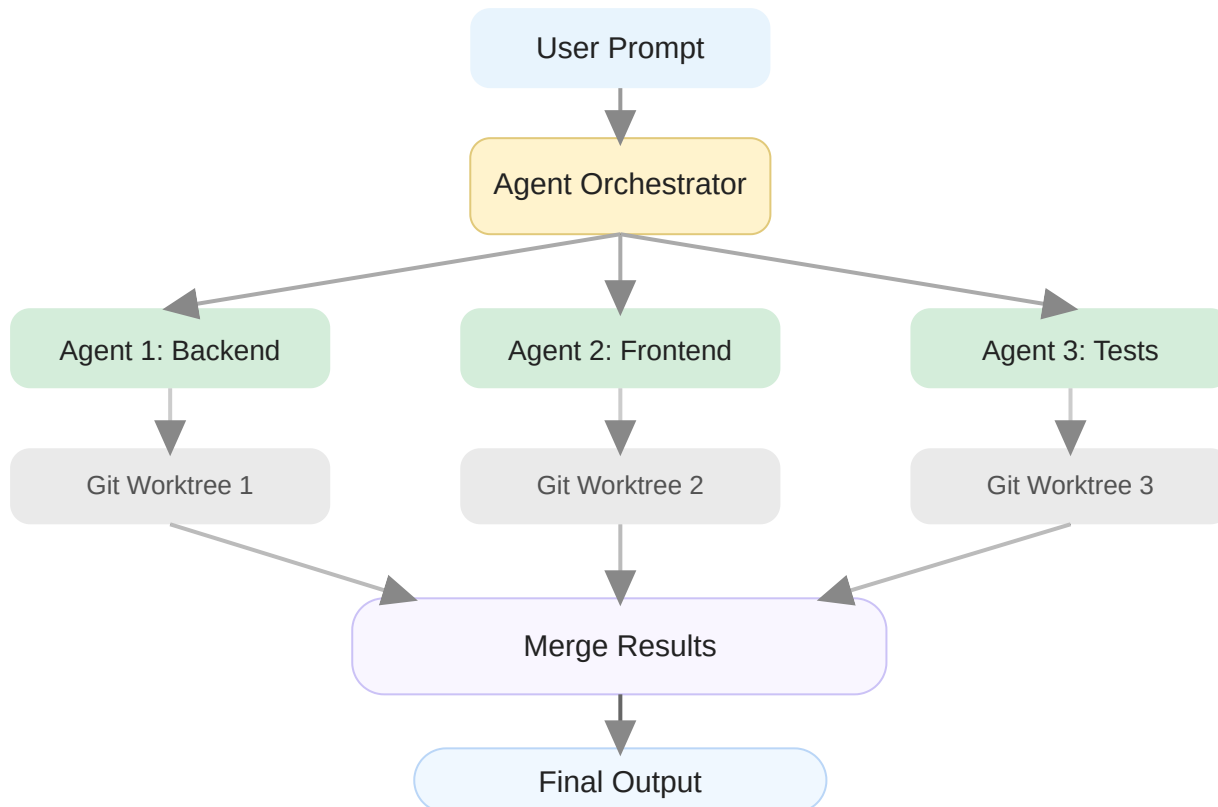
When to Use

- Recurring mistakes
- Critical security patterns
- Team consistency

Watch Out

- Can slow you down
- Keep under 500 lines
- Better context > more rules

Multi-Agent Architecture



Each agent operates in isolated git worktree → No conflicts

Model Context Protocol (MCP)

Open protocol connecting AI to tools & data

Like USB for AI—standardized interface

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Tools

Functions AI executes



Prompts

Templated workflows



Resources

Structured data sources

GitHub MCP

- Create PRs
- Review code
- Manage issues

```
npx @modelcontextprotocol/server-github
```

Figma MCP

- Extract design tokens
- Generate components
- Sync UI updates

```
bunx cursor-talk-to-figma-mcp
```

GitMCP

- Turn any repo into MCP
- Replace github.com → gitmcp.io
- Instant context access

```
gitmcp.io/facebook/react
```

Database MCPs

- PostgreSQL
- SQLite
- Query and inspect

```
@modelcontextprotocol/server-postgres
```

Demo: Cursor Configuration Setup



See: `presentation/scripts.md`

Demo 1: Setting Up Cursor Configuration

Questions?

Open floor—ask about anything from this section

Break

10 Minutes 

Building Your First Full App with Cursor

AI-Driven Project Initialization



1. Describe

Problem, users,
features in plain
English



2. Define

Tech stack, database,
auth, deployment



3. Generate

Let Cursor scaffold
everything

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The Context File Approach

- Create docs/context.md with your vision
- Reference it inside prompts: 'Generate the project structure for @docs/context.md '

Example: Context File

DeepWork AI - Focus Productivity App

Problem

Developers struggle with multitasking and distractions.
Need a tool to prioritize tasks and maintain deep work focus.

Users

Software engineers, designers, product managers

Core Features

1. Task management with priority levels
2. Focus timer for single-task concentration
3. AI chat to quickly add tasks via natural language

Tech Stack

- Frontend: Next.js 15+ (App Router), TypeScript, Tailwind
- Backend: Supabase (database + auth)
- AI: OpenAI API for chat
- Deployment: Vercel

Database Schema

[Let Cursor generate based on features]

ChatGPT + Cursor Workflow

1 ChatGPT

Planning Phase:

- Brainstorm features
- Structure ideas
- Create context.md
- Plan architecture



2 Cursor

Implementation:

- Generate code
- Multi-file operations
- Debug errors
- Refactor



Use ChatGPT (or Claude/Gemini/etc.) for exploration, Cursor for implementation

Demo: Building Your First Full App

Setting Up Supabase Backend

1. Create Project

- Visit supabase.com
- Create new project
- Copy API URL & anon key

2. Environment Variables

```
NEXT_PUBLIC_SUPABASE_URL=...  
NEXT_PUBLIC_SUPABASE_ANON_KEY=...
```

3. Prompt Cursor

```
@docs/context.md
```

```
Install Supabase client and create  
the database schema for tasks with:  
title, description, priority,  
deadline, completed status, user_id.
```

 **Cursor generates SQL migrations
+ RLS policies**

Creating UI Components

Prompt for Component Generation:

Create the following components following our project rules:

1. TaskCard - Display single task with priority badge
2. TaskList - Grid of TaskCards with filtering
3. AddTaskModal - Form to create new tasks
4. FocusTimer - Countdown timer for focused work

Use Tailwind, make them responsive, and include TypeScript interfaces for all props.

Iterative Development Loop

1. Generate code with Agent or Composer



2. Run the app - Check for errors: `npm run dev`



3. Copy error messages - Paste into Cursor chat



4. Let Cursor fix - Review the changes



5. Commit - Save working state with git

Testing with AI Assistance

Generate Test Suites:

@src/components/TaskCard.tsx

Create a complete test suite using Jest and React Testing Library.

Include:

- Rendering tests
- User interaction tests
- Edge cases
- Accessibility tests

Testing with AI Assistance

Generate Test Suites:

```
@src/components/TaskCard.tsx
```

```
Create a complete test suite using Jest and React Testing Library.
```

```
Include:
```

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- Edge cases
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Component Tests



Integration Tests



E2E Tests

Questions?

Open floor—ask about anything from this section

Break

10 Minutes 

Advanced Architecture Patterns



Atomic Design

- **Atoms** - Button, Input
- **Molecules** - SearchBar
- **Organisms** - Header
- **Templates** - Layouts
- **Pages** - Full views



State Management

- **Local** - useState
- **Global Client** - Zustand
- **Server** - React Query
- **URL** - searchParams



Custom Hooks

- Single responsibility
- Named exports
- TypeScript interfaces
- Reusable logic



Performance

- React.memo()
- useMemo()
- useCallback()
- Code splitting

Demo: AI-Powered Quiz App

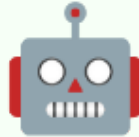
Quiz App: Tech Stack

AI-Generated Quiz Application



Frontend

Next.js 15+
TypeScript
Tailwind CSS



AI

OpenAI API
Instructor
Structured Output



Backend

Supabase
Real-time
Row Level Security

Architecture Diagrams with Mermaid

Prompt Cursor to Generate Diagrams:

Create a Mermaid diagram showing the data flow in our quiz app:

- User selects topic

- API generates questions with OpenAI

- Questions displayed one at a time

- Answers stored in Supabase

- Results calculated and shown

Architecture Diagrams with Mermaid

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✓ Diagram Types

- flowchart - Logic flow
- sequenceDiagram - Interactions
- classDiagram - Data structures

💡 Best Practice

Start small, layer upward, multiple perspectives

[1][Cursor Cookbook - Mermaid Diagrams](#)

Figma MCP Integration



Figma Design

- Design tokens
- Component specs
- Typography



React Code

- Pixel-perfect
- Tailwind classes
- Responsive

Prompt Example:

```
@figma file:quiz-app-designs
```

Generate the QuestionCard component matching the Figma design.
Extract all design tokens (colors, spacing, shadows) and
create a matching React component with Tailwind.

[1] [Cursor Talk to Figma MCP](#)

Questions?

Open floor—ask about anything from this section

Full-Stack Application

Backend API Development

Prompt for Complete REST API:

Create a complete REST API for our quiz app:

Endpoints:

- POST /api/quiz/generate - Create new quiz
- GET /api/quiz/[id] - Get quiz by ID
- POST /api/quiz/[id]/answer - Submit answer
- GET /api/quiz/[id]/results - Get final results
- GET /api/quiz/history - User's past quizzes

Include: Authentication, validation (Zod), error handling, rate limiting, API documentation comments

Backend API Development



Auth



Validation



Rate Limiting

Database Optimization

Optimization Strategies

- **Indexes** - Speed up queries
- **JOINS** - Reduce round trips
- **Pagination** - Limit results
- **Caching** - Redis layer
- **Connection Pooling** - Supervisor

Prompt Cursor:

```
@src/lib/db/queries.ts
```

Optimize this query for performance:

- Add proper indexes
- Use JOIN instead of multiple queries
- Implement pagination
- Add caching strategy

State Management & Real-time



Zustand Store

Quiz state, answers, timer



Optimistic Updates

Instant UI feedback



Real-time Sync

Supabase subscriptions

Optimistic Update Pattern

1. Update UI immediately (optimistic)
2. Send request to server
3. If error: rollback UI changes
4. If success: keep UI as-is

Result: App feels instant ⚡

Demo: Test, Iterate, Deploy

Questions?

Open floor—ask about anything from this section

Break

10 Minutes 

Deployment & Production

Deployment with Vercel

From Code to Production in 10 Minutes



Zero Config

Auto optimizations



Edge Network

Global CDN



Instant Rollback

One-click revert

CI/CD Pipeline

Generate GitHub Actions Workflow:

Create a GitHub Actions workflow for our Next.js app:

On every PR:

1. TypeScript type checking
2. ESLint
3. Tests (unit + integration)
4. Build
5. Deploy preview to Vercel

On merge to main:

All above + E2E tests + Production deploy + Slack notification

CI/CD Pipeline

 **Type Check**

 **Test**

 **Deploy**

Production Readiness Checklist

Security

- Env vars secured
- SSL + CORS configured
- Rate limiting enabled
- Dependency & secret scanning

Performance

- CDN + edge caching
- Image optimization
- Code splitting
- Database indexes

Code Quality

- Type safety (strict TS)
- ESLint + a11y checks
- 80% test coverage

Monitoring & Recovery

- Error tracking (Sentry)
- Uptime + analytics
- Automated backups
- Rollback strategy

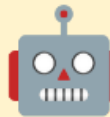
Automated PR Workflow

Staff Engineer Tip: Handle PR Feedback Fast



1. Screenshot

Capture PR
comments



2. Cursor Plan

Drag into Cursor,
generate plan



3. Execute

Fix items one by one

Even faster: run `/review` (`/review-bugbot`, `/review-security`) before you push—
Bugbot integrates with GitHub & GitLab PRs

Demo: Vercel & Github Actions

Cost Optimization Strategies



Current Costs

- Vercel Pro: \$20/month
- Supabase Pro: \$25/month
- OpenAI API: ~\$50/month
- Upstash Redis: \$10/month

Total: ~\$105/month



Optimizations

- ☒ Switch to DeepSeek (10x cheaper)
- ☒ Aggressive caching
- ☒ Free tier for previews
- ☒ Optimize DB queries
- ☒ Rate limiting

New Total: ~\$40/month



Prompt Cursor for Cost Analysis

Analyze our app's costs and suggest optimizations to reduce spending while maintaining performance.

Key Takeaways

✓ Context is King

Strategic @-mentions, clean project structure

✓ Three-Step Framework

Explore → Plan → Build

✓ Production-Ready

Testing, CI/CD, monitoring, security

✓ MCP Integration

GitHub, Figma, databases



Remember

Cursor augments your workflow—it doesn't replace engineering judgment. Use it to eliminate tedious work and focus on what matters.

Additional Resources



Documentation

- [Cursor Docs](#)
- [Changelog](#)
- [Composer 2.5](#)



MCP Servers

- [Official Servers](#)
- [GitMCP](#)
- [Figma MCP](#)



Videos

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